ARAZVERA TECHNOLOGIES / ENGINEERING CAPABILITY CATALOG / NOV 2026

LEVEL 04

Enterprise Resilience & Security Engineering

Acceptance-tested backend reliability, durable recovery and conservative release control

VERSION 0.4.0 — Public engineering showcase

The underlying system is a **working, disposable, single-host laboratory**, not a customer's production deployment, third-party certification or uptime guarantee.

Engineering proposition

Business-critical backends can fail at boundaries between HTTP APIs, identity, tenant isolation, transaction processing, workers, operational monitoring and releases. ARAZVERA Level 4 supports scoped assessments and implementation with measured synthetic acceptance, conservative change control and documented evidence.

Specialist service capabilities

Engagement family	Practical scope
Architecture and Reliability Audit	Service boundaries, failure pathways, operational risk, SLO gaps and reviewable acceptance criteria
Multi-Tenant Security & Identity	PostgreSQL FORCE RLS, restricted roles, RS256/JWKS validation, replay protection and operator role design
Transaction and Worker Recovery	Transactional Outbox, bounded retry, deduplication, fenced leases, dead letters and durable delivery receipts
Safe Staging Releases	Two real synthetic HTTP backends, 200-request measured Canary, HOLD, stable routing and explicit rollback
Incident Readiness	Real health probes, persistent incident transitions, atomic internal alert jobs and independently confirmed receipts
Backend Reliability Remediation	Approved Python/FastAPI/PostgreSQL/Docker changes with negative tests, runbooks and handover

Customer-oriented deliverables

Technical findings, isolated reproduction, approved corrective engineering, automated regression evidence, release and rollback rules, documented security boundaries and English handover. All production actions and third-party costs require separate customer authorization.

Verified technical evidence

Latest private source commit: 339125a1c3f9c8030bbc534a1fd5f8db73aec45d
Private CI run: 42793494053 — **four of four GitHub Actions jobs successful**.

Acceptance scope	Actual observed result
Complete isolated Docker Pytest suite	**129 passed, zero failed**, two non-blocking cache warnings
Local Python unit suite	**102 passed, zero failed**
Real Chrome/Selenium dashboard	**13 of 13** interactions passed
Synthetic Canary and rollback	**3 of 3**, including faulty candidate HOLD and Stable restoration
API / PostgreSQL scoped outage	**17 of 17** recovery and durability checks
Separate PostgreSQL logical restore	**8 of 8** integrity and RLS checks
Independent Worker stop and resume	**7 of 7**, queued event delivered after restart
Operator role authorization	Viewer, operator and admin JWT scope checks with actual internal signed JWKS verification
Transactional incident notification	Atomic OPEN/RESOLVED events, duplicate prevention, real internal HTTP dispatch and database receipts

Do not add the group totals: several suites contain overlapping tests. No customer production performance, real SSO integration, independent penetration test or external alert delivery is asserted.

Laboratory architecture

The disposable Docker project includes a FastAPI Control Plane, restricted PostgreSQL 16 application account and tenant roles, independent Worker, periodic observer, synthetic downstream receiver, ephemeral RS256 identity service and two separate Stable/Candidate backends.

A fixed internal candidate receives 200 real measured HTTP samples. The decision, error status and latency evidence are recorded in PostgreSQL. Failures produce HOLD and restore the synthetic Stable route. Incidents atomically queue outbox events; the independent Worker records delivery receipts only after internal acknowledgement.

Operator controls support an opt-in signed-only laboratory JWT mode with viewer, operator and admin roles. The regular local token mode remains available. Neither represents a deployed customer identity provider.

Engagement model and production gates

Scope-driven project delivery

Stage	Engineering handover
1 — Assess	Written architecture, ownership, access authorization, risk register and acceptance goals
2 — Reproduce	Bounded failure experiment and measured baseline in approved resources
3 — Engineer	Reviewed source changes with tests for negative, duplicate and partial-failure cases
4 — Validate	Recovery, permission, canary and durability evidence; documented unknown outcomes
5 — Handover	English runbook, explicit release/rollback policy, source review and acceptance register

Engagements may use contractual milestones or agreed time-and-materials. Infrastructure and third-party accounts remain client-owned. Rates and commercial outcomes are proposal-specific, not guaranteed.

Explicitly NOT yet production-certified

- Corporate IAM/SSO, MFA, identity lifecycle, security audit or customer authorization have not been integrated.
- Independent-host failover, encrypted offsite DR and measured customer RPO/RTO have not been verified.
- Synthetic Canary and rollback do **not** publish or revert real customer application images.
- Lab alert delivery does **not** notify real email/SMS/messaging/paging/webhook recipients.
- SHA-256 checksums are not digital signatures; local scanners and CI are not independent certification.
- The whole Windows/Docker Engine reboot path has not been tested; only named lab container recovery controls are verified.
- No private source, passwords, customer data, third-party production systems or unrelated AERON/NEXA resources are included.

Company and next steps

ARAZVERA TECHNOLOGIES
Public Level 4 engineering repository
ARAZVERA GitHub organization

Request a written, acceptance-based engineering assessment using a sanitized architecture and a reproducible failure scenario.

This catalog communicates verified laboratory capability and scope limitations, not previous Enterprise customer contracts or a promise of production readiness.